DIFFUSION MODELS

1. Overview

How do we approximately sample from the data distribution P_{data} given the datapoints $x_1, \dots, x_n$?

[Luo22]:

Consider a distribution P with density $p(x) \propto \exp(-V(x))$ supported on $\mathbb{R}^D$. Here V is called the potential function and $\nabla \log p = \nabla V$ is known as the score. The simplest way to sample from P is MCMC. Construct the Langevin diffusion $dX_t = -\nabla V(X_t)dt + \sqrt{2}dB_t$ for a BM B, which has stationary distribution p, and run many sample paths for a sufficiently large amount of time.

Thus if we had access to the score of P, we could sample from P. However, real data distributions are often not supported on all of the ambient $\mathbb{R}^D$, so the score is undefined. One way to think about diffusion models is that they convolve P with gaussians at multiple scales. For any $t > 0$, the convolution P_t will be supported on all of $\mathbb{R}^D$ and the score will be well-defined. A diffusion model learns the score of P_t for multiple values of t. (There is some theory on low dimensional structure, see [CHZW23]...)

[CMFW24]:

A diffusion model has a forwards process (encoder) and backwards process (decoder). The forward process is an O-U diffusion, like $dX_t = -\frac{1}{2}X_t dt + dW_t$. Let P_t denote the distribution of X_t at time t, and let p_t denote its density. Under an O-U diffusion, the signal decays exponentially. The forward process is run until $P_T \approx N(0, I_d)$. The backwards process is a Langevin diffusion, like $dX_t = (\frac{1}{2}X_t + \nabla \log p_{T-t}(X_t))dt + d\overline{W}_t$. The backwards process starts with a completely noisy image and denoises it in such a way that at time t, it is distributed as P_{T-t}. In particular, at time T the distribution of the backwards process is P_{data}. (Add derivation of backwards process...)

So a natural generative algorithm would be to sample gaussian noise and run the backwards process from that initial condition. The issue is that we don't know the score $\nabla \log p_t(x)$. A diffusion model solves this issue by fitting a neural network $\hat{s}(t, x)$ to the score $\nabla \log p_t(x)$ at each time t; this is "score matching". The most intuitive loss function would be

$$\int_0^T w(t) E_{x_t \sim P_t}[|\nabla \log p_t(x_t) - s(t, x_t)|^2]dt \tag{1}$$

for some weight function w. Since we don't know $\nabla \log p_t$, this is infeasible. Instead, add conditioning on X_0 to reduce to the known transition probabilities:

$$\int_0^T w(t) E_{x_0 \sim P_0}[E_{x_t \sim P_t|P_0}[|\nabla \log \phi_t(x_t|x_0) - s(t, x_t)|^2]] dt \tag{2}$$

Finally, replace P_0 with the empirical measure on the datapoints.

Early stopping time: when the data is supported on a low dimensional subspace of the ambient $\mathbb{R}^D$, the score blows up as $t \to 0$. Note that P_t is convolution of P_{data} with a gaussian and thus has support $\mathbb{R}^D$. However P_{data} might be singular with respect to the Lebesgue measure on $\mathbb{R}^D$, in which case the score of P_t for $t \approx 0$ can become very large. Thus the lower limit of the score matching objective is typically replaced by some $t_0 > 0$.

U-Net, how it is trained, add...

Approximation theory of the score function: difficulty of the extra time dimension. [CHZW23] reduces it to standard NN approximation theory, add...

Statistical question of whether the distribution of generated data is close to P_{data}. It turns out that it is sufficient to perform score matching, add...

Diffusion models come in two types: unconditional and conditional. Conditional models require that the model samples from the data distribution conditioned on some information $Y = y$ (like a prompt for DALL-E). The forward and backwards processes generalize to $X|Y = y$ just as before. But now we must learn the conditional score $\nabla \log p_t(x|y)$ for each time t. To do so, we apply Bayes' rule: $\nabla \log p_t(x|y) = \nabla \log p_t(x) + \nabla \log p_t(y|x)$. The first term on the RHS is the unconditional score. The second term $p_t(y|x)$ is a probability prediction that x has label y, so it is a classifier. We can train a classifier c_t and combine it with an unconditional score to obtain $\nabla \log p_t(x|y)$ for all t. However, for $t \approx T$, the performance of a classifier will necessarily be poor. An alternative is classifier-free guidance, which trains a conditional and unconditional score simultaneously by randomly ignoring the conditional information. (Classifier guidance is still good for fine-tuning.)

Aside from image generation, applications to graphical models, RL, optimization, add...

Less theory for conditional models and effect of guidance?, less theory for low dimensional data?, add...

2. Bayesian interpretation

[Luo22]:

Many generative models can be understood in a Bayesian framework. A Bayesian model consists of observable data x and latent variables z with a joint distribution $p(x, z)$. Observations are generated by first drawing z from the prior $p(z)$, and then drawing x from the sampling distribution $p(x|z)$ (abuse of notation). We want to learn the posterior distribution $p(z|x)$ of z after observing the data x, which by Bayes rule equals $\frac{p(x|z)p(z)}{\int p(x|y)p(y)dy}$.

Unfortunately, the posterior can be hard to compute. In Variational Bayes, one approximates $p(z|x)$ in KL divergence by a distribution $q_\phi(z|x)$ in some family:

$$\text{argmin}_\phi KL(q_\phi(z|x)||p(z|x)) \tag{3}$$

It turns out that the Evidence Lower Bound (ELBO) is a more convenient objective function:

$$ELBO = E_{q_\phi(z|x)}[\log \frac{p(x,z)}{q_\phi(z|x)}] \tag{4}$$

It can be shown that $ELBO = \log p(x) - KL(q_\phi(z|x)||p(z|x))$, so that maximizing the ELBO is equivalent to minimizing the KL. (Write $p(x,z) = p(z|x)p(x)$, split the log to get $-KL + E[\log p(x)] = -KL + \log p(x)$, since the log likelihood doesn't depend on z.)

If we apply Bayes rule in the opposite manner (i.e. $p(x,z) = p(x|z)p(z)$), we can rewrite the ELBO as

$$ELBO = E_{q_\phi(z|x)}[\log p(x|z)] - KL(q_\phi(z|x)||p(z)) \tag{5}$$

The second term is called the "prior matching term". We don't want the learned posterior to be too far from our prior on z. The first term is called the "reconstruction term", in analogy to the reconstruction error of a vanilla AE. If z is sampled from the true posterior $p(z|x)$, then by definition we expect the likelihood of z given x to be large. In Variational Bayes, we might also need to learn the sampling distribution $p(x|z)$, so we use the family $p_\theta(x|z)$.

A variational autoencoder (VAE) performs Variational Bayes by optimizing the ELBO with a NN. The prior $p(z)$ is taken to be $N(0,I)$, the posterior $q_\phi(z|x)$ is selected from a family of gaussians $N(\mu_\phi(x), \sigma^2_\phi(x)I)$, and $x|z$ is a deterministic function?, add... Given a sample $\{x_1, \ldots, x_n\}$, the empirical ELBO is

$$\text{argmax}_{\phi,\theta} \sum_{i=1}^{n} ((\frac{1}{m}\sum_{j=1}^{m} \log p(x_i|z_j^{(i)})) - KL(q_\phi(z|x_i)||p(z))) \tag{6}$$

Here $\{z_1^{(i)}, \ldots, z_m^{(i)}\}$ are sampled from $q_\phi(z|x_i)$ for each i to estimate the reconstruction term.

To optimize this via GD, we need to take gradients with respect to ϕ, θ. We can't treat the $z_j^{(i)}$ as constants, because their distributions depend on ϕ. Thus we reparametrize $q_\phi(z|x)$ as $z = \mu_\phi(x) + \sigma_\phi(x) \odot \varepsilon$ for $\varepsilon \sim N(0,I)$. Rewriting the objective function in terms of these ε's allows us to do GD.

The architecture of the NN has two parts: the encoder and the decoder. The encoder represents the posterior $q_\phi(z|x)$; the "encoding" of x is the posterior. The gaussian posterior can be parametrized by its mean and variance $(\mu_\phi(x), \sigma^2_\phi(x))$. The decoder represents the sampling distribution $p_\theta(x|z)$: "decoding" a latent z yields its sampling distribution over x. Once the VAE is trained, we can sample from $p(x)$ by drawing $z \sim p(z)$ and applying the decoder to z.

We now discuss Hierarchical VAE (HVAE) models. Instead of using a single latent variable, a Hierarchical Bayes model utilizes multiple latent variables $z_1, \dots, z_T$. The hierarchy is such that $z_{i+1:T} := z_{i+1}, \dots, z_T$ determine the sampling distribution of z_i for all i. That is, z_T is sampled, then z_{T-1} is sampled from $p(z_{T-1}|z_T)$, and so on, until the observable x is sampled from $p(x|z_{1:T})$. For an HVAE the ELBO becomes

$$ELBO = E_{q_\phi(z_{1:T}|x)}[\log \frac{p(x, z_{1:T})}{q_\phi(z_{1:T}|x)}] \tag{7}$$

In a Markovian HVAE, the sequence $z_T, z_{T-1}, \dots, z_1, x$ is a Markov chain.

Finally, we may define a Variational Diffusion Model (VDM) by mapping our notion of a diffusion model onto the framework of a Markovian HVAE. A VDM is a Markovian HVAE where for each i, the posterior $z_i|x$ is a noisy version of x obtained by running an O-U process. We choose T such that $z_T|x$ is approximately $N(0, I)$. We abuse notation and denote z_i by x_i for $1 \le i \le T$. The specific amount of noise is obtained by discretizing the O-U process. Therefore we have specified the encoder, and we only need to learn the sampling distribution decoder. The ELBO (which in this case is equivalent to log likelihood?) can be written as

$$\begin{aligned} ELBO = {} & E_{q(x_1|x_0)}[\log p_\theta(x_0|x_1)] - KL(q(x_T|x_0)||p(x_T)) && (8) \\ & - \sum_{t=2}^{T} E_{q(x_t|x_0)}[KL(q(x_{t-1}|x_t, x_0)||p_\theta(x_{t-1}|x_t))] && (9) \end{aligned}$$

(Abusing notation by using q.) The first term is the reconstruction term and the second is a prior matching term ($\approx$ zero). Each term in the sum is a "denoising matching term". $p_\theta(x_{t-1}|x_t)$ is our guess of the distribution of x_{t-1} given the noisier x_t; we try to undo one step of noising. $q(x_{t-1}|x_t, x_0)$ is the true distribution of x_{t-1} given x_t and the clean x_0; we try to undo one step of noising knowing the clean datapoint. We want the KL to be as small as possible. It turns out that $q(x_{t-1}|x_t, x_0)$ is a gaussian, so we parametrize $p_\theta(x_{t-1}|x_t)$ by a gaussian and match its mean and variance. We can exactly match the variance $\sigma_q^2(t)I$, because it doesn't depend on x_0. However, the mean $\mu_q(x_t, x_0)$ of $q(x_{t-1}|x_t, x_0)$ is a linear combination of both x_0 and x_t. Computing the mean $\mu_\theta(t, x_t)$ of $p_\theta(x_{t-1}|x_t)$ therefore reduces to estimating x_0 by a NN $\hat{x}_\theta(t, x_t)$.

We see that a VDM aims to estimate x_0 given the noisy x_t for any t; if it can do this, it can learn $p(x_{t-1}|x_t)$ for all t, and it can sample. (Equivalently, the VDM aims to estimate x_{t-1} given x_t for each t?...) But we can reparametrize to obtain the score-matching interpretation we began with. Tweedie's formula states that if $z \sim N(\mu, \Sigma)$, where μ is a random variable and Σ is known, then we have

$$E[\mu|z] = z + \Sigma \nabla_z \log p(z) \tag{10}$$

We now apply Tweedie's formula to $z = x_t$. By construction, $x_t \sim N(\sqrt{c_t}x_0, (1 - c_t)I)$. We obtain

$$E[\sqrt{c_t}x_0|x_t] = E[\mu_{x_t}|x_t] = x_t + (1 - c_t)\nabla_{x_t} \log p(x_t) \tag{11}$$

So estimating x_0 is equivalent to estimating the score $\nabla \log p(x_t)$.

(If you had a DM with $T = 1$ latent, then Tweedie says $\nabla \log p_{data}$ reduces to computing the conditional mean $E[x|x+\text{lots of noise}]$. But inverting lots of noise is hard, so instead repeatedly invert a small amount of noise. This is why we set T large.)

3. Stochastic localization

[fAS22] (based on "Algorithmic stochastic localization" paper [EAMS22]):

How do we sample from a distribution μ? Here, we consider the Sherrington-Kirkpatrick model, which is the mean-field Gibbs measure with Hamiltonian $\sum G_{ij}\sigma_i\sigma_j$, where $G \sim GOE(n)$.

The simplest way is MCMC. For spin systems (Ising, SK), one can use Glauber dynamics, which samples σ_i from the conditional distribution $p(\cdot|(\sigma_j)_{j\neq i})$. We expect sampling to be easy in the high temperature regime $\beta < 1$, when the correlations are weak. (Why is $\beta_c = 1$?) The rate of convergence is quantified by a Poincaré inequality for the Markov process. The Poincaré constant is related to the spectral diameter of G. However, linear time mixing for Glauber has only been proved for $\beta < 1/4$ (Bauerschmidt-Bodineau '19).

This paper presents an algorithm based on stochastic localization (SL). SL was introduced by Eldan as a proof technique, but El Alaoui-Montanari-Sellke converted it into a sampling algorithm for the SK model.

One version of SL is as follows. Suppose we want to sample from μ, but our only access to μ is through paths of the following BM with drift: $y_t = tx^* + B_t$, where $x^* \sim \mu$ and B is BM. If we can sample from the law of y, can we sample from μ? Intuitively, the answer should be yes, because the SNR is increasing over time. Since $\frac{y_t}{t} \xrightarrow{as} x^*$, the quantity $\frac{y_t}{t}$ should approximately be distributed as μ for sufficiently large t. (We start to see a connection to diffusion models. To sample from a distribution, we instead sample a particular diffusion and try to denoise it. However, the parametrization is reverse; in a diffusion model, SNR decreases over time.) Because we don't know x^*, we should instead describe y according to the "annealed dynamic" $dy_t = m_t dt + dW_t$, where $m_t = E[x^*|\mathcal{F}_t] = E[x^*|y_t]$ is the posterior mean of the drift at time t. The full posterior distribution of x^* given $\mathcal{F}_t$ is the random measure $\mu_t(dx) \propto \exp(\langle y_t, x\rangle - \frac{t|x|^2}{2})\mu(dx)$. (This is because the likelihood $p(y_t|x)$ is given by $\exp(-\frac{|y_t - tx|^2}{2t})$.)

As time goes on, μ_t concentrates (due to the $-\frac{t|x|^2}{2}$) and converges to the random measure δ_{m_∞} with mean $m_\infty \sim \mu$. Eldan originally defined SL as the measure-valued process $(\mu_t)_{t\geq 0}$. In the context of the SK model, there is a physical meaning to everything: μ_t is the SK measure with an external magnetic field y_t that evolves according to the annealed dynamic, and its mean m_t is the magnetization.

This leads to SL sampling. Run the diffusion y for a large amount of time t. Compute the posterior mean $m_t = E[x^*|y_t]$. Return m_t. Since $m_\infty \sim \mu$, we should have that m_t is approximately distributed as μ. (It might seem like SL sampling is the same as MCMC, because both rely on running a Markov process. But the difference lies in the information that we condition on. In MCMC, we sample σ_i by revealing information about the marginal of μ at all other sites. In SL, we sample σ by revealing information about the convolution of μ with a heat kernel, which involves all sites.)

In general, the difficulty in implementing SL sampling lies in computing m_t. In the specific case of the SK model, we can overcome this difficulty because the magnetization m obeys a fixed-point equation called the TAP equations. The TAP equations are obtained by minimizing the nonconvex TAP free energy F_{TAP} as a function of m.

To find m_t, there are two stages. First, run AMP iterations. This yields an approximate solution $\hat{m}_t^{(k)}$ that lands in a region of strong convexity of F_{TAP}. Second, run gradient descent from $\hat{m}_t^{(k)}$ to find the true m_t.

Of course, we do this for each value of t. The resulting algorithm has complexity $O(n^2)$, and the distribution μ^{alg} of the resulting draw is close to μ in Wasserstein distance.

[Mon23]: denoising diffusion (the "time reversal approach") isn't equivalent to SL sampling? (Though by Tweedie, the time reversal approach is also computing a conditional mean.)

The processes are different. In the time reversal approach, the distribution of the backwards process Y_t converges to the deterministic measure μ. In the SL approach, the (non-annealed) process Y_t is such that the distribution of Y_t/t converges to the random measure δ_{x^*}. Thus we output Y_t/t for large t. Since Y_t solves the annealed SDE, we could also output $m_t(Y_t, t)$.

One can generalize SL beyond the isotropic gaussian process $Y_t = tx^* + B_t$ defined above. An observation process Y can be any process such that Y_t grows more informative about x over time. Y and x don't even have to live in the same space. Y should be chosen based on μ, so that the conditional mean m_t is easy to compute. Examples: erasure process, linear observation process (which requires an extra step).

Empirical observations: learn $m(y,t)$ by a NN approximation $\hat{m}_\theta(y,t)$. In the isotropic gaussian case, given training data (x_i) the objective becomes $\frac{1}{n}\sum |x_i - \hat{m}_\theta(tx_i + \sqrt{t}G; t)|^2$. Once you have $\hat{m}_\theta$, you can run the annealed SDE. The architecture of the NN matters. In section 6 of [Mon23], μ is a distribution on images with long-range correlations. Using an isotropic gaussian process and fitting m with a CNN does badly at sampling, because a CNN learns short-range correlations in the distribution of x. In this case, a fix is to augment x by the global average of its pixel values and to use a higher-d linear observation process. (Rigorous result in section 7.)

Q: what are other global features of distributions that occur in practice/in stat mech models? How can we design NNs to capture them?

Calculations:

SL observation process: $dY_t = m(Y_t, t)dt + dB_t$

Define: $\bar{Y}_t := \frac{1}{\sqrt{t(1+t)}} Y_t$

Ito product rule: $d\bar{Y}_t = \frac{-(1+2t)}{2(t(1+t))^{3/2}} Y_t dt + \frac{1}{\sqrt{t(1+t)}} dY_t$

$= \frac{-(1+2t)}{2t(1+t)} \bar{Y}_t dt + \frac{1}{\sqrt{t(1+t)}} m(Y_t, t)dt + \frac{1}{\sqrt{t(1+t)}} dB_t$

$= \frac{-(1+2t)}{2t(1+t)} \bar{Y}_t dt + \frac{1}{\sqrt{t(1+t)}} m(\sqrt{t(1+t)}\bar{Y}_t, t)dt + \frac{1}{\sqrt{t(1+t)}} dB_t$

Claim: this is the *time-changed* DM backward process. (First term has typo in [Mon23]...) Thus running SL sampling is equivalent to sampling DM backwards process.

Why? Rewriting the backward process:

Attempt #1: (abusing the notation $\bar{Y}$) try the backward process with naive time change $t = T - s$ defined only for $t \in [0, T]$. (Here s is time parameter for forward process X.)

$d\bar{Y}_t = (\bar{Y}_t + 2\nabla \log q_{T-t}(\bar{Y}_t))dt + \sqrt{2} dB_t$

Apply Tweedie to replace score with a posterior mean. Tweedie in full generality: given $y = x + z$, where the noise z is $N(0, \Sigma)$, our best guess of x is an adjustment of y: $E[x|y] = y + \Sigma\nabla \log q(y)$, where q is pdf of y. I.e. $\nabla \log q_t(y) = \Sigma^{-1}(E[x|y] - y)$.

Here we seek $\nabla \log q_t$, so we consider $X_t \sim N(e^{-t}X_0, (1 - e^{-2t})I)$. Obtain $\nabla \log q_t(y) = \frac{1}{1-e^{-2t}}(E[e^{-t}X_0|X_t = y] - y)$.

Drift term becomes:

$\bar{Y}_t + 2\frac{1}{1-e^{-2(T-t)}}(E[e^{-(T-t)}X_0|X_{T-t} = \bar{Y}_t] - \bar{Y}_t)$

$= -\frac{1+e^{-2(T-t)}}{1-e^{-2(T-t)}} \bar{Y}_t + \frac{2e^{-(T-t)}}{1-e^{-2(T-t)}} E[X_0|X_{T-t} = \bar{Y}_t]$

Want to rewrite in terms of the function m given by $m(y, t) := E[X|tX + \sqrt{t}G = y]$, where $X \sim \mu$ and $G \sim N(0, I)$. Stuck; we have $E[X_0|e^{-(T-t)}X_0 + \sqrt{1 - e^{-2(T-t)}}G = y]$. SNR increases at the wrong rate in t. Need a time change...

Attempt #2: Claim: given time change $t(s)$, define $\bar{Y}$ via $d\bar{Y}_t = \bar{F}(t, X_t)dt + \sqrt{\bar{g}(t)} d\bar{B}_t$, where $\bar{F}(t, y) = (y + 2\nabla \log q_{s(t)}(y))|s'(t)|$ and $\bar{g}(t) = 2|s'(t)|$. Then $\bar{Y}_t \overset{d}{=} X_{s(t)}$ for all t.

Take $t(s) = \frac{1}{e^{2s}-1}$. (Why? This equals $\frac{e^{-2s}}{1-e^{-2s}}$, the SNR. So we reparametrize to make SNR of backward process increase linearly.) Note that we set $T = \infty$.

$s(t) = \frac{1}{2}\log(1+t^{-1})$.

$|s'(t)| = |\frac{1}{2}\frac{1}{1+t^{-1}}(-t^{-2})| = \frac{1}{2t(1+t)}$

The MG part is clearly $\frac{1}{\sqrt{t(1+t)}}dB_t$.

Score: $\nabla \log q_{s(t)}(y) = \frac{1}{1-e^{-2s}}(E[e^{-s}X_0|e^{-s}X_0 + \sqrt{1-e^{-2s}}G = y] - y)$

$e^{2s} = 1 + t^{-1}$

$e^{-2s} = \frac{1}{1+t^{-1}}$

$e^{-s} = \frac{1}{\sqrt{1+t^{-1}}} = \frac{\sqrt{t}}{\sqrt{1+t}}$

$1 - e^{-2s} = \frac{t^{-1}}{1+t^{-1}} = \frac{1}{1+t}$

$\sqrt{1-e^{-2s}} = \frac{1}{\sqrt{1+t}}$

$E[X_0|\frac{\sqrt{t}}{\sqrt{1+t}}X_0 + \frac{1}{\sqrt{1+t}}G = y]$

$= E[X_0|tX_0 + \sqrt{t}G = \sqrt{t(1+t)}y]$

$= m(\sqrt{t(1+t)}y, t)$

So the score equals $(1+t)(\frac{\sqrt{t}}{\sqrt{1+t}}m(\sqrt{t(1+t)}y,t) - y)$.

So our drift is $\bar{F}(t,y) = (y + 2(score)) \cdot \frac{1}{2t(1+t)}$

$= \frac{1}{2t(1+t)}((-1-2t))y + 2\sqrt{t(1+t)}m(\sqrt{t(1+t)}y,t))$

$= -\frac{1+2t}{2t(1+t)}y + \frac{1}{\sqrt{t(1+t)}}m(\sqrt{t(1+t)}y,t)$

We obtain $d\bar{Y}_t = -\frac{1+2t}{2t(1+t)}\bar{Y}_t dt + \frac{1}{\sqrt{t(1+t)}}m(\sqrt{t(1+t)}\bar{Y}_t, t)dt + \frac{1}{\sqrt{t(1+t)}}dB_t$ with stationary measure μ, as desired.

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SDE satisfied by observation process Y:

$Y_t := tx^* + W_t$

Why $dY_t = m(Y_t, t)dt + dB_t$? (Recall this is essential for actual sampling.)

Lipster-Shiryaev, Thm 7.12: can we represent a general Ito process ξ given by $d\xi_t = \beta_t(\omega)dt + dW_t$, $\xi_0 = 0$ as a diffusion process? (Recall a diffusion process is stronger; the drift must ξ-adapted, i.e. $\beta_t(\omega) = \alpha_t(\xi(\omega))$.) Yes, we can do this on $[0,T]$ so long as $\int_0^T E[|\beta_t|] < \infty$. Let $\alpha_t(\xi) = E[\beta_t|\mathcal{F}_t^\xi]$ be the conditional mean of β_t. Then $B_t = \xi_t - \int_0^t \alpha_s(\xi)ds$ is a $(\mathbb{F}^\xi, P)$-BM and $d\xi_t = \alpha_t(\xi) + dB_t$.

For us: Y in Ito form is $dY_t = \beta_t dt + dW_t$, where $\beta_t = x^*$ (not ξ-adapted). For any finite T, $E[|x^*|^2] < \infty$ implies $\int_0^T E[|x^*|] < \infty$, thus Thm 7.12 applies on any $[0,T]$. We set $\alpha_t = E[x^*|\mathcal{F}_t^Y] = E[x^*|Y_t] =: m(Y_t, t)$.

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Second-order Tweedie identity from [BBDD24]:

Suppose we observe the noisy $y = x + z$, where x has density q_0 and $z \sim N(0, \sigma^2 I)$.

y has density q given by the convolution

$q(y) = \int q_0(x) \frac{1}{(2\pi\sigma^2)^{d/2}} \exp(-\frac{1}{2\sigma^2}|y-x|^2)dx$

Score:

$\nabla \log q(y) = \frac{1}{q(y)} \int q_0(x) \frac{1}{(2\pi\sigma^2)^{d/2}} \exp(-\frac{1}{2\sigma^2}|y-x|^2) \cdot \frac{-(y-x)}{\sigma^2} dx$

(separate the y and x) $= \frac{1}{\sigma^2} E[x|y] - \frac{y}{\sigma^2}$

Once rearranged, this yields Tweedie: $E[x|y] = y + \sigma^2 \nabla \log q(y)$.

But let's take another derivative to obtain hessian:

$\int \frac{1}{q(y)} q_0(x) \frac{1}{(2\pi\sigma^2)^{d/2}} \exp(-\frac{1}{2\sigma^2}|y-x|^2) \frac{x}{\sigma^2} dx - \frac{y}{\sigma^2}$

Second term: $-\frac{1}{\sigma^2} I$

First term: recall that if f is real-valued, $\nabla_y f(y)v = v \nabla f(y)^T$.

So let's compute grad of $f(y) = \frac{1}{q(y)} q_0(x) \frac{1}{(2\pi\sigma^2)^{d/2}} \exp(-\frac{1}{2\sigma^2}|y-x|^2)$ by product rule.

Note $\nabla f(y) = \nabla \log f(y) \cdot f(y)$; keep the density $f(y)$ intact.

$\nabla \log f(y)^T = \nabla \log \frac{1}{q(y)}^T + \frac{1}{\sigma^2}(x-y)^T = -(\nabla \log q(y)^T + \frac{1}{\sigma^2} y^T) + \frac{1}{\sigma^2} x^T$ (using Tweedie) $= -\frac{1}{\sigma^2} E[x|y]^T + \frac{1}{\sigma^2} x^T$

So $\int f(y) \cdot \frac{x}{\sigma^2} \nabla \log f(y)^T dx$

$= -\frac{1}{\sigma^4} E[x|y] E[x|y]^T + \frac{1}{\sigma^4} E[xx^T|y]$

$= \frac{1}{\sigma^4} \Sigma$

Combining, we obtain $\nabla^2 \log q(y) = \frac{1}{\sigma^4} \Sigma - \frac{1}{\sigma^2} I$.

How do these read for DMs? Under O-U, $x_t = e^{-t}x_0 + \sqrt{1-e^{-2t}}g$. So let $y \leftarrow x_t$, $x \leftarrow e^{-t}x_0$, and $z \leftarrow \sigma_t g$, where $\sigma_t = \sqrt{1-e^{-2t}}$. (The e^{-t} in front of x_0 fudges things a bit.)

Tweedie: $\nabla \log q_t(x_t) = \frac{1}{\sigma_t^2}e^{-t}m_t - \frac{1}{\sigma_t^2}x_t$.

Second order Tweedie: $\nabla^2 \log q_t(x_t) = \frac{1}{\sigma_t^4}e^{-2t}\Sigma_t - \frac{1}{\sigma_t^2}I$.

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Consider the isotropic gaussian SL process. Eldan's theorem is a recurrence for the densities of the μ_t's.

Eldan's thm: if $L_t = d\mu_t/d\mu$, then there exists a Y-adapted BM W such that $dL_t(x) = L_t(x)\langle x - a_t, dW_t\rangle$, where $L_0 \equiv 1$ and where $a_t = E[x^*|y_t]$ is the mean of μ_t.

(This is called "linear tilt" localization, because $L_{t+\delta}(x) \approx L_t(x)(1 + \langle x - a_t, \Delta W_t\rangle)$; at each step, we multiply the density by a (random) linear function with expected slope 0.)

Pf: At first glance, it's not clear how we construct W. It turns out to be the BM W such that $dy_t = a_t dt + dW_t$.

Due to Ito, the desired is not quite $d\log L_t(x) = MG$, but it's close. We know $L_t(x) = \frac{1}{Z_t}\exp(\langle y_t, x\rangle - \frac{t}{2}|x|^2)$, so $\log L_t(x) = \langle y_t, x\rangle - \frac{t}{2}|x|^2 - \log Z_t$. (Don't forget PF.)

$d\log L_t(x) = \langle x, dy_t\rangle - \frac{1}{2}|x|^2 dt - \log Z_t$

Lipster-Shiryaev: $dy_t = a_t dt + dW_t$ (this is the W).

$Z_t = \int \exp(h_t(x))\mu(dx)$ (a SP)

Ito: $dZ_t = \int [\exp(h_t(x))dh_t(x) + \frac{1}{2}\exp(h_t(x))d\langle h_t(x)\rangle]\mu(dx)$

$dh_t(x) = \langle x, dy_t\rangle - \frac{1}{2}|x|^2 dt$

$d\langle h_t(x)\rangle = |x|^2 dt$

So $dZ_t = \int \langle x, dy_t\rangle \exp(h_t(x))\mu(dx)$

$= Z_t\langle a_t, dy_t\rangle$

So $d\langle Z\rangle_t = |\int x\exp(h_t(x))\mu(dx)|^2 dt$

$= Z_t^2 a_t^2 dt$

Ito: $d\log Z_t = \frac{1}{Z_t}dZ_t - \frac{1}{2}\frac{1}{Z_t^2}d\langle Z\rangle_t$

$= \langle a_t, dy_t\rangle - \frac{1}{2}a_t^2 dt$

Combining, $d \log L_t(x) = \langle x - a_t, dy_t \rangle - \frac{1}{2}(|x|^2 - a_t^2)dt$

Use the drift to kill the drift: $\langle x - a_t, dW_t \rangle + [\langle x - a_t, a_t \rangle - \frac{1}{2}(|x|^2 - a_t)^2]dt$

$= \langle x - a_t, dW_t \rangle - \frac{1}{2}|x - a_t|^2 dt$

Note $d\langle \log L_t(x) \rangle = |x - a_t|^2 dt$

Finishing touches: we want $dL_t(x)$, so use Ito to compute $d \exp(that)$:

$dL_t(x) = L_t(x) d \log L_t(x) + \frac{1}{2} L_t(x) d\langle \log L_t(x) \rangle$

$= L_t(x) \langle x - a_t, dW_t \rangle$

as desired.

We can use Eldan's thm to write down ODEs for the evolution of the SL covariance matrix A_t. Claim: $\frac{d}{dt} E[A_t] = -E[A_t^2]$...

Translate to DMs...

(Q: we do SL sampling by estimating a_t; is there ever any benefit to estimating A_t?...)

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Less related to SL as mechanism to produce spectral gap lower bounds via approximate conservation of variance...

4. Fine-tuning

[UZB+24]: entropy-regularized fine tuning via stochastic control. We follow the explanation in [Tan24]. Suppose you have a pretrained DM with sampling distribution p_{pre}, and you want to fine tune it according to a reward function r. The function r is learned from preference data. In order to avoid overfitting the fine tuned model to r, we consider an entropy regularized objective:

$$\text{argmin}_{p_{ft}} E_{Y \sim p_{ft}}[r(Y)] - \alpha KL(p_{ft} || p_{pre}) \tag{12}$$

The first term maximizes the reward, and the second term forces p_{ft} to stay close to p_{pre}. Using the Gibbs variational principle, the optimum is $p_{ft}(y) \propto e^{r(y)/\alpha} p_{pre}(y)$, a tilted version of p_{pre} that places more weight where r is larger.

This provides us with a formula for p_{ft} in terms of p_{pre}, but leaves us no way to sample from p_{ft}; all we have is a sampler from p_{pre}! It would be nice if we could modify the backwards sampling process such that at time T, the distribution of Y_T is not p_{pre} but rather p_{ft}. We can formulate this as a stochastic control problem. Abusing notation, if our original backwards SDE is $dY_t = b_t dt + \sigma_t dB_t$ with $Y_0 \sim p_{noise}$, we instead consider $dY_t = (b_t + u_t)dt + \sigma_t dB_t$ with $Y_0 \sim \nu$. We must choose the initial distribution ν and the control u in order for $Y_T \sim p_{ft}$. An objective that achieves this aim is

$$\text{argmin}_{u,\nu} E_{Q^{u,\nu}}[r(Y_T)] - \alpha KL(Q^{u,\nu}||Q) \tag{13}$$

where $Q^{u,\nu}$ denotes the law of the controlled SDE over path space, and $Q = Q^{0,p_{noise}}$ denotes the law of the original SDE. To solve this, you solve ν given u^* (this is just Gibbs variational principle), and then you solve u^* via a corresponding HJB equation.

One caveat: now you also need to sample from ν^* to initialize the backwards SDE! To do you can write down a simpler SDE running from time $-T$ to 0 such that at time 0, $Y_0 \sim \nu^*$. Then proceed to time T by running the controlled SDE above.

(Stochastic control is a particularly good framework here because we're modifying an existing SDE in order to sample differently. But for models that don't use diffusion, RL approaches might be more straightforward...)

(There are challenges in learning r, because it converts a collection of preferences to a quantitative reward function. If not done carefully, you can face "reward collapse": [SCLS23].)

5. Papers

[CCL+22]: Given L^2 accurate scores, how many steps must the discretized SDE take to come close to P_{data} in TV distance? Intuitive argument goes via Girsanov. Consider the law $Q_T^{\leftarrow}$ of the true reverse process, and consider the law P_T of the SGM initialized according to q_T. (It's easy to swap out the true initialization $N(0, I)$ with q_T by properties of O-U.) It suffices to bound $KL(Q_T^{\leftarrow}||P_T)$. Recall that $Q_T^{\leftarrow}, P_T$ are laws of processes that agree up to their drift. Girsanov tells us that the change of measure on path space is given by $\frac{dP_T}{dQ_T^{\leftarrow}} = \exp(\int_0^T b_s dB_s - \frac{1}{2}\int_0^T |b_s|^2 ds)$, where b is the difference in the drifts of the true reverse process and the SGM (i.e. $\sqrt{2}$ times the difference between true and estimated scores), and where B is the driving BM of the true reverse process. (Why not $\frac{dQ_T^{\leftarrow}}{dP_T}$? Because we want the integrand in the next line to be a $Q_T^{\leftarrow}$-GM.) Thus $KL(Q_T^{\leftarrow}||P_T) = \frac{1}{2}\int_0^T E_{Q_T^{\leftarrow}}[|b_s|^2]ds$ (the SI dies because it's a $Q_T^{\leftarrow}$-MG), which equals $\sum_k \int_{t_k}^{t_{k+1}} E[|s_{t_k}(Y_{t_k}) - \nabla \log q_{T-s}(Y_s)|^2]ds$. From here, L^2 accurate scores and Lipschitzness lets us conclude. In reality, we must modify this proof due to the failure of the Novikov condition; apply Girsanov to a stopped process...

[BBDD24]: An improvement on the above iteration complexity: linear in dimension d, without Lipschitz assumption on the score. Recall that backward process Y is equivalent to SL process under a time change... They achieve better control of discretization error using an identity from SL about the means/covariances of the SL process.

Lemma 1: if (m_t, Σ_t) denote posterior mean of forward process X, and if $\sigma_t^2 = 1 - e^{-2t}$ is noise level at time t, then $\frac{\sigma_t^3}{2\dot{\sigma}_t}\frac{d}{dt}E[\Sigma_t] = E[\Sigma_t^2]$.

Discretization error: recall that we simulate the SDE by using the estimated score at t_k for the whole duration of the interval $[t_k, t_{k+1}]$. Thus we care about L^2 distance between $\nabla \log q_{T-t}(Y_t)$ and (an estimated version of) $\nabla \log q_{T-t_k}(Y_{T-t_k})$. Lemma 2 gets a better bound on this L^2 error integrated over t. How? Study the general difference $E_{s,t} = |\nabla \log q_{T-t}(Y_t) - \nabla \log q_{T-s}(Y_s)|^2$. (If the score was Lipschitz, we could bound this easily by adding/subtracting $\nabla \log q_{T-t}(Y_s)$, say.) By Ito, we can use the dynamic of Y_t to get the dynamic of $\nabla \log q_{T-t}(Y_t)$. Recall Ito isometry: $E[(dZ)^2] = E[d\langle Z\rangle]$. Apply this to $Z_t = \nabla \log q_{T-t}(Y_t) - \nabla \log q_{T-s}(Y_s)$ to obtain a formula for $\frac{d}{dt}E_{t,s}$. Ultimately we separate t, s to obtain $\frac{d}{dt}E_{t,s} \leq E[|\nabla \log q_{T-t}(Y_t)|^2] + 2E[|\nabla^2 \log q_{T-s}(Y_s)|^2]$. By Tweedie, $\nabla \log q_{T-t}(Y_t)$ and $\nabla^2 \log q_{T-s}(Y_s)$ can be expressed in terms of m, Σ. Using Lemma 1, we get a nice upper bound purely in terms of σ's and $\mathrm{tr}\Sigma_{T-t}$. Once we integrate E_{t,t_k} over each interval and sum, we get the discretization bound in Lemma 2.

[GPPX23]: Interested in sample complexity, not iteration complexity. Given a NN function class, how many samples to learn accurate score estimate? Turns out L^2 score condition is too restrictive. Relax to an L^2 score condition on $1-\delta$ of the space; then *polylog*$(1/\gamma)$ samples are possible to learn γ-smoothed data distribution. This modified score condition suffices to perform sampling; simple modification of [CCL+22]?.

Why is relaxed L^2 score condition learnable in *polylog*$(1/\gamma)$? Want to show that the ERM is close to the true score s^*. The ER is $\hat{E}_{x,z}[\tilde{l}(s,x,z)]$, where $\tilde{l} = |s(x) - s^*(x)|^2 + 2\langle s(x) - s^*(x), \Delta\rangle$. Show that if s is far from s^*, it yields large ER whp. New metric: s is "bad" if $P(|s(x) - s^*(x)| > \varepsilon/\sigma) \geq \delta$. To analyze $\hat{E}_{x,z}[\tilde{l}]$, first approximate by $\hat{E}_x[\min(E_{z|x}[\tilde{l}], C)]$ for some constant C. Note that $E_{z|x}[\tilde{l}] = |s(x) - s^*(x)|^2$. Call the resulting expression A_x. When s is "bad", A_x is the average of a bunch of iid rvs, each of which takes on the value ε^2/σ^2 at least δ of the time. So whp, $A_x \gtrsim \delta\varepsilon^2/\delta^2$. (This assumes we have $m \gtrsim 1/\delta$ samples, of course.) Next, justify the approximation by showing that $\hat{E}_{x,z}[\tilde{l}] \geq A_x/2$ whp. Use concentration to handle terms $\hat{E}_{z|x=x_i}[\tilde{l}(s,x_i,z)]$ with x_i such that $|s(x_i) - s^*(x_i)|^2$ is big/small (?). (This requires $m > O(\textit{desired})$...) This implies that each "bad" s yields a large ER wp $\geq 1-\delta$, and we can union bound by discretizing the NN class to finish.

Why is L^2 score condition too restrictive?

Example 1: there exists a pair of distributions p_1, p_2 such that given m samples, (1) you can't tell which distribution the sample came from, but (2) the L^2 distance between the scores is large (under either p_1 or p_2). (I.e. imagine p_1 is the truth; you incorrectly guess p_2 50% of the time, and incur a large L^2 error.) Let $p_i = (1-\eta)N(0,\sigma^2) + \eta N(\pm R, \sigma^2)$. Then you need $m \sim 1/\eta$ samples to distinguish them; otherwise, both p_1, p_2 look like $N(0,\sigma^2)$. But the L^2 error in the score under the true p_1 is $\sim R^2\eta/\sigma^4$. (The scores differ by R/σ^2 near $+R$, and the probability of that region is $\sim \eta$.) (Here R stands for $1/\gamma$, i.e. we need $\geq$ *poly*$(1/\gamma)$ samples?) (I.e. if you sample by first picking a p that seems likely to have generated the data, then you need $\geq$ *poly*$(1/\gamma)$ samples to choose correctly.)

Example 2: there exists a pair of distributions p_1, p_2 such that given m samples from p_1, the score of p_2 is L^2 close to the score of p_1 with respect to empirical distribution, but far with respect to $L^2(p_1)$. (I.e. if you sample by picking a p that gives small empirical L^2 error, then you need $\geq poly(1/\gamma)$ samples to ensure a small L^2 population error.)

[MW23]: Implement diffusion sampling for spiked Wigner model for $\beta > \beta_*$. Phrased as SL sampling with gaussian observation process: input is a denoiser $m(y,t)$ to estimate the conditional mean. Use AMP estimator of m. This builds on the work of Montanari-Venkatramanan '21 on AMP for low-rank estimation. (Also, in general, given an accurate denoiser, they present discretization error estimates for SL sampling.) (Note: posterior mean estimator for SL process is equivalent to a score estimator for DM reverse process?...)

[GDKZ24]: a natural Q: how do DM/AR/flow generative models compare to MCMC?

Recall that DM/AR/flow sampling relies on computation of some posterior mean. For DM/flow, requires denoising additive gaussian noise. For AR, requires denoising a randomly masked input. (Flow models: interpolate $x \sim P_{data}$ and $z \sim N(0, I)$ via the path $y(t) = \alpha(t)x + \beta(t)z$. To generate a new sample, run a backwards ODE starting from z with velocity $-E[\dot{\alpha}(t)x + \dot{\beta}(t)|y(t) = y]$. This reduces to computing $E[x|\alpha(t)x + \beta(t)z]$.)

So we are asking: given a perfect denoiser, how does denoising-based sampling compare to MCMC? (Whether DNNs learn denoisers well is a separate Q.)

Answer: each wins in a certain regime. (I.e. sampling can be easier than denoising.) (Q: this answers a question from a meeting, and leads us to ask: is MCMC the best we can hope for in certain cases?...) Key computation: phase diagrams via free entropy calculation.

Where MCMC wins: models with 1-RSB at high-temp $T \in (T_d, T_{tri})$. Let P_{data} be a p-spin glass model at temp T. MCMC phase diagram is well-understood: efficient for $T > T_d$. The posterior for gaussian noise at SNR γ is the tilted P_γ. (Can also study the posterior for randomly masked noise, the pinned P_θ.) We seek the phase diagram for denoising in (T, γ). We replace P_γ with a planted model (?) and compute the free entropy Φ_{RS} as a function of the overlap order parameter χ (I thought this is 1-RSB?...). At some values (T, γ), there is a unique maximizer of the free entropy (?). However, at other (T, γ) there are two maximizers; this corresponds to coexistence of two distinct Gibbs measures (?). (Intuitively, your algo might converge to the wrong value of χ?...) This coexistence makes sampling hard in the "red region" located to the left of $T = T_{tri}$. But for denoising-based sampling to work at a given T, we need to denoise at all $\gamma \geq 0$. So denoising fails for $T < T_{tri}$. (Q: an idea suggested in the paper is to consider more complex noising procedures in the hope that T_{tri} decreases...)

Where denoising wins: models with stat-to-comp gap, like rank-1 matrix estimation at high SNR. Here P_{data} is the posterior on the hidden vector. MCMC can sample for noise smaller than Δ_{MCMC}, but denoising via AMP can sample up to $\Delta_{alg} > \Delta_{MCMC}$. (Attempted justification: denoising only looks at low-d marginals, while MCMC moves in the configuration space?...) (Q: it's not intuitive to me what the difference between these models is...)

(Q: do these results give us heuristics so that for real Bayesian problems, we can decide between MCMC and generative models?...)

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